

12.01 Linear-Quadratic Systems

Common problems and solution methods

Rule guide: recognize the task first

Recognize	Apply these rules
Functions And Models	Read the function form and the context before choosing a graphing or modeling method.
The expression contains the chapter's key notation or model.	Write the known quantities, choose the matching formula, calculate carefully, and check the result in context.

Example 1: Identify the model

Problem

$$f(x) = x^2$$

How to solve it

Write the known quantities and restrictions.

$$m = \frac{y_2 - y_1}{x_2 - x_1}$$

Check the result against the original statement.

Example 2: Apply the key rule

Problem

$$y = 2(x - 3)^2 - 5$$

How to solve it

Write the known quantities and restrictions.

$$\text{vertex of } y = a(x - h)^2 + k : (h, k)$$

Check the result against the original statement.

Example 3: Check the result

Problem

$$\frac{f(4) - f(1)}{4 - 1}$$

How to solve it

Write the known quantities and restrictions.

$$\text{average rate} = \frac{f(b) - f(a)}{b - a}$$

Check the result against the original statement.